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Inventor(s)

YEN; HSIAO-TSUNG et al.

TRANSFORMER DEVICE

Abstract

A transformer device includes first, second, and third coils. The first coil includes first segments and at least one first connecting portion. The second coil includes second segments and second connecting portions. The third coil, including third segments and third connecting portions, couples the first and second coils. Part of the third segments are coupled in parallel through the third connecting portions, and at least one part of the first segments and at least one part of the second segments are arranged between those third segments. The first segments and the at least one first connecting portion form a first winding, the second segments and the second connecting portions form a second winding, and the third coil and innermost and adjacent turns in each of the first and the second windings do not form crossing portions in two opposite sides of the third coil.

Inventors: YEN; HSIAO-TSUNG (Hsinchu, TW), WANG; PO-CHIH (Hsinchu, TW)

Applicant: REALTEK SEMICONDUCTOR CORPORATION (HSINCHU, TW)

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] The present application is a divisional application of U.S. application Ser. No. 17/512,696 filed Oct. 28, 2021, which claims priority to U.S. provisional application Ser. No. 63/115,073 filed Nov. 18, 2020, and also claims priority to Taiwan Application Serial Number 110111936 filed Mar. 31, 2021, the disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] The present disclosure relates to a transformer device. More particularly, the present disclosure relates to a planar transformer device for power combination.

2. Description of Related Art

[0003] Certain integrated circuits (ICs) for radio frequency signals have to convert signals between a common mode and a differential mode. A BALUN is usually utilized in this kind of signal conversion. A BALUN is one of many applications of a transformer and is implemented with coil(s) in the ICs. Therefore, a good design for coils in terms of excellent coupling, higher quality factor, and improved line balancing becomes more and more significant.

SUMMARY

[0004] In some aspects, a transformer device includes a first coil, a second coil, and a third coil. The first coil includes a plurality of first segments and at least one first connecting portion, in which the plurality of first segments are coupled to each other through the at least one first connecting portion. The second coil includes a plurality of second segments and a plurality of second connecting portions, wherein the plurality of second segments are coupled to each other through the plurality of second connecting portions. The third coil is configured to couple the first coil and the second coil. The third coil includes a plurality of third segments and a plurality of third connecting portions, a part of the plurality of third segments are coupled in parallel with each other through the plurality of third connecting portions, and at least one part of the plurality of first segments and at least one part of the plurality of second segments are arranged between the part of the plurality of third segments. The plurality of first segments and the at least one first connecting portion form a first winding, the plurality of second segments and the plurality of second connecting portions form a second winding, and the third coil and an innermost turn and a turn adjacent to the innermost turn in each of the first winding and the second winding do not form a plurality of crossing portions in two opposite sides of the third coil.

[0005] In some aspects, a transformer device includes a first coil, a second coil, and a third coil. The first coil includes a plurality of first segments and at least one first connecting portion, in which the plurality of first segments are coupled to each other through the at least one first connecting portion. The second coil includes a plurality of second segments and a plurality of second connecting portions, in which the plurality of second segments are coupled to each other through the plurality of second connecting portions. The third coil is configured to couple the first coil and the second coil, in which the third coil comprises a plurality of third segments and a plurality of third connecting portions, a part of the plurality of third segments are coupled in parallel with each other through the plurality of third connecting portions, and at least one part of

the plurality of first segments and at least one part of the plurality of second segments are arranged between the part of the plurality of third segments. The plurality of first segments and the at least one first connecting portion form a winding having a first turn and a second turn, and the plurality of first segments and the at least one first connecting portion sequentially form, from a first terminal of the first coil to a second terminal of the first coil, a quarter of the first turn, a quarter of the second turn, a quarter of the first turn, a half of the second turn, a quarter of the first turn, a quarter of the second turn, and a quarter of the first turn.

[0006] These and other objectives of the present disclosure will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description that are illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1A is a schematic diagram of a transformer device according to some embodiments of the present disclosure.

[0008] FIG. 1B is a schematic diagram of two coils in FIG. 1A according to some embodiments of the present disclosure.

[0009] FIG. 1C is a schematic diagram of a coil in FIG. 1A according to some embodiments of the present disclosure.

[0010] FIG. 2 is a schematic diagram of a transformer device according to some embodiments of the present disclosure.

[0011] FIG. 3 is a schematic diagram of a transformer device according to some embodiments of the present disclosure.

[0012] FIG. 4 is a schematic diagram of a transformer device according to some embodiments of the present disclosure.

[0013] FIG. 5 is a schematic diagram of a transformer device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0014] The terms used in this specification generally have their ordinary meanings in the art and in the specific context where each term is used. The use of examples in this specification, including examples of any terms discussed herein, is illustrative only, and in no way limits the scope and meaning of the disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given in this specification.

[0015] As used herein, “about” or “substantially” shall generally mean within 20 percent, preferably within 10 percent, and more preferably within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “about” or “substantially” can be inferred if not expressly stated.

[0016] Further, for ease of description, spatially relative terms, such as “left,” “right,” “lower,” “upper,” “below,” “over,” and the like, may be used herein to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0017] In this document, the term “coupled” may also be termed as “electrically coupled,” and the term “connected” may be termed as “electrically connected.” “Coupled” and “connected” may mean “directly coupled” and “directly connected” respectively, or “indirectly coupled” and “indirectly connected” respectively. “Coupled” and “connected” may also be used to indicate that

two or more elements cooperate or interact with each other. In this document, the term “circuit” may indicate an object, which is formed with one or more transistors and/or one or more active/passive elements based on a specific arrangement, for processing signals.

[0018] As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Although the terms “first,” “second,” etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the embodiments. For ease of understanding, like elements in various figures are designated with the same reference number.

[0019] Reference is made to FIG. 1A to FIG. 1C, FIG. 1A is a schematic diagram of a transformer device **100** according to some embodiments of the present disclosure, FIG. 1B is a schematic diagram of a coil **120** and a coil **140** in FIG. 1A according to some embodiments of the present disclosure, and FIG. 1C is a schematic diagram of the coil **160** in FIG. 1A according to some embodiments of the present disclosure. In some embodiments, the transformer device **100** may operate (but not limited to) as a power combiner, which may couple signals from two symmetrical coils to a single coil, in order to output a single signal. For ease of understanding, components in the transformer device **100** are separately shown in FIG. 1B and FIG. 1C. In some embodiments, the transformer device **100** is formed with the coils **120** and **140** in FIG. 1B and the coil **160** in FIG. 1C.

[0020] Each of the coil **120** and the coil **140** may be a planar coil. As shown in FIG. 1B, the coil **120** includes segments L1 (shown in horizontal stripes) and connecting portion CP1 (shown in diagonal stripes). The segments L1 are coupled to each other through the connecting portions CP1, in order to form the coil **120**. For example, at least one via (shown in black) is arranged on each of two terminals of each connecting portion CP1, in order to couple the connection portion CP1 to the corresponding segment L1. Similarly, the coil **140** includes segments L2 (shown in white) and connecting portions CP2 (shown in diagonal stripes). The segments L2 are coupled to each other through the connecting portions CP2, in order to form the coil **140**. For example, at least one via is arranged on each of two terminals of each connecting portion CP2, in order to couple the connecting portion CP2 to the corresponding segment L2.

[0021] In some embodiments, each of the coils **120** and **140** may operate as a differential inductor. For example, the coil **120** includes a terminal P11 and a terminal P12. The terminal P11 and the terminal P12 may output (or receive) a set of differential signals. In some embodiments, a middle terminal (not shown) between the terminal P11 and the terminal P12 may be configured to receive a common mode voltage (e.g., an AC ground voltage). A wire length between the terminal P11 and the middle terminal is substantially the same as that between the terminal P12 and the middle terminal. In some embodiments, the middle terminal may operate as a center tap terminal. Similarly, the coil **140** includes a terminal P21 and a terminal P22. The terminal P21 and the terminal P22 may output (or receive) a set of differential signals. In some embodiments, a middle terminal between the terminal P21 and the terminal P22 may be configured to receive a common mode voltage (e.g., an AC ground voltage). A wire length between the terminal P21 and the terminal is substantially the same as that between the terminal P22 and the middle terminal. In some embodiments, the middle terminal may operate as a center tap terminal.

[0022] In some embodiments, a reference line A-A' is presented at a central location between the terminal P11 and the terminal P12, such that the coil **120** and the coil **140** are substantially mirror images of each other with respect to the reference line A-A'. It is understood that, the reference line A-A' is substantially arranged at a central location between the coil **120** and the coil **140**, and the reference line A-A' is not a physical component in the transformer device **100**.

[0023] The coil **160** may be a planar inductor, which is configured to couple the coil **120** and the coil **140**. As shown in FIG. 1C, the coil **160** includes segments L3 (shown in dots) and connecting

portions CP3 (shown in diagonal stripes). The segments L3 are coupled to each other through the connecting portions CP3, in order to form a ring structure. For example, at least one via is arranged on each of two terminals of each connecting portion CP3, in order to couple the connecting portion CP3 to the corresponding segments L3. The coil **160** may be applied to (but not limited to) a single-ended signaling. For example, the coil **160** includes a terminal P31 and a terminal P32. The terminal P31 may receive a single-end signal, and the terminal P32 may receive a DC voltage (e.g., a common mode voltage or a ground voltage). In some embodiments, the ring structure of the coil **160** has a mirror image symmetry with respect to the reference line A-A'. In other words, as shown in FIG. 1A, the transformer device **100** substantially has a bilateral symmetry structure. In some embodiments, a middle terminal between the terminal P31 and the terminal P32 may be configured to receive a common mode voltage (e.g., an AC ground voltage). A wire length between the terminal P31 and the middle terminal is substantially the same as that between the terminal P32 and the middle terminal. In some embodiments, the middle terminal may operate as a center tap terminal.

[0024] As shown in FIG. 1A or FIG. 1B, the segments L1 and the connecting portions CP1 form a first winding having four turns. For example, from the terminal P11 to the terminal P12, the segments L1 and the connecting portions CP1 sequentially form about a quarter of a first turn (e.g., an outermost turn), about a quarter of a second turn, about a quarter of the first turn, about a quarter of the second turn, about a quarter of a third turn, about a quarter of a fourth turn (e.g., an innermost turn), about a quarter of the third turn, about a half of the fourth turn, about a quarter of the third turn, about a quarter of the fourth turn, about a quarter of third turn, about a quarter of the second turn, about a quarter of the first turn, about a quarter of the second turn, and about a quarter of the first turn. Similarly, the segments L2 and the connecting portions CP2 form a second winding having four turns. For example, from the terminal P21 to the terminal P22, the segments L2 and the connecting portions CP2 sequentially form about a quarter of a first turn (e.g., an outermost turn), about a quarter of a second turn, about a quarter of the first turn, about a quarter of the second turn, about a quarter of a third turn, about a quarter of a fourth turn (e.g., an innermost turn), about a quarter of the third turn, about a half of the fourth turn, about a quarter of the third turn, about a quarter of the fourth turn, about a quarter of third turn, about a quarter of the second turn, about a quarter of the first turn, about a quarter of the second turn, and about a quarter of the first turn.

[0025] As mentioned below, the segments L1, L2, L3, the connecting portions CP1, CP2, and CP3 forms crossing portions in two opposite sides of the coil **160**. In some embodiments, these crossing portions are formed by each turn in the first winding and the second winding and the coil **160** in the two opposite sides. In some embodiments, the two opposites may be a first side and a second side, the first side is a side along the direction of $-X$, and the second side is a side along the direction of $+X$. In some embodiments, the terminals P31 and P32 are arranged in a third side (e.g., a side along the direction of $+Y$) of the coil **160**, and the terminals P11, P12, P21, and P22 are arranged in a fourth side (e.g., a side along the direction of $-Y$) of the coil **160**, in which the third side and the fourth side are different from the first side and the second side. In other words, the crossing portions and the terminal P11, P12, P21, P22, P31 or P32 are arranged in different sides. Moreover, the crossing portions in the first side are symmetrical to the crossing portions in the second side.

[0026] In FIG. 1A, in the first side (e.g., the side along the direction of $-X$) of the coil **160**, each of the coils **120** and **140** crosses the coil **160** four times. For example, in the first side, two connecting portions CP1 and CP2 cross the segments L3 in the outermost turn of the coil **160**, and another two connecting portions CP1 and CP2 cross the segments L3 in the third turn of the coil **160**. Moreover, in the first side, the corresponding segments L1 and L2 cross the connecting portion CP3 in the outermost turn of the coil **160**, and another corresponding segments L1 and L2 cross the connecting portion CP3 in the third turn of the coil **160**. Based on the symmetrical structure, in the second side (e.g., the side along the direction of $+X$) of the coil **160**, each of the coil **120** and the coil **140**

crosses the coil **160** four times as well.

[0027] Similarly, in the third side (e.g., the side along the direction of +Y) of the coil **160**, each of the coil **120** and the coil **140** crosses the coil **160** four times. In the fourth side (e.g., the side along the direction of -Y) of the coil **160**, each of the coil **120** and the coil **140** crosses the coil **160** six times. In some embodiments, the first side and the second side are opposite sides of the coil **160** (e.g., a left side and a right side of the coil **160**), and the third side and the fourth side are opposite sides of the coil **160** (e.g., a upside and a downside of the coil **160**). In greater detail, in the third side, the segments **L1** in the outermost turn of the coil **120** and the segments **L2** in the outermost turn of the coil **140** cross the connecting portions **CP3** between the first turn and the second turn of the coil **160**. In the third side, the segment **L1** in the third turn of the coil **120** and the segment **L2** in the fourth turn of the coil **140** cross the connecting portion **CP3** between the third turn and the fourth turn of the coil **160**, and the connecting portion **CP1** between the third turn and the fourth turn of the coil **120** and the connecting portion **CP2** between the third turn and the fourth turn of the coil **140** cross the segment **L3** in the third turn of the coil **160**. Similarly, in the fourth side, the segments **L1** in the first turn of the coil **120** and the segments **L2** in the first turn of the coil **140** cross the connecting portions **CP3** between the first turn and the second turn of the coil **160**, and the segment **L1** in the second turn of the coil **120** and the segment **L2** in the second turn of the coil **140** cross the connecting portion **CP3** between the second turn and the third turn of the coil **160**. In the fourth side, the segments **L1** in the third turn of the coil **120** and the segments **L2** in the third turn of the coil **140** cross the connecting portions **CP3** between the third turn and the fourth turn of the coil **160**, and the connecting portions **CP1** between the third turn and the fourth turn of the coil **120** and the connecting portions **CP2** between the third turn and the fourth turn of the coil **140** cross the segment **L3** in the third turn of the coil **160**.

[0028] With the above arrangement, the coils **120**, **140**, and **160** may form crossing portions via the connection portions **CP1**, **CP2**, and **CP3**, respectively. According to different purposes of circuit applications, a direction of the current on the connecting portions **CP1** and that on the connection portions **CP2** may be set to be the same or different. An inductance value of the connecting portions **CP1** and that of the connecting portions **CP2** may be changed significantly according to different current directions, and thus the current direction may be controlled to meet different purposes of circuit applications. Equivalently, the segments **L1**, **L2**, and **L3** may be twisted around each other via the connecting portions **CP1**, **CP2**, and **CP3**. With such arrangements of the embodiment, both of the connecting portions **CP1** and **CP2** are able to uniformly couple with the connecting portions **CP3**.

[0029] In some embodiments, a part of the segments **L3** are coupled in parallel with each other via the connecting portions **CP3**, and at least one part in the segments **L1** and at least one part in the segments **L2** are arranged between the part of the segments **L3**. For example, as shown in FIG. 1C, the segments **L3** in a first turn (e.g., the outermost turn) and a second turn of the coil **160** are coupled in parallel with each other via the connecting portions **CP3**. The segments **L3** in a third turn and a fourth turn (e.g., an inner turn) of the coil **160** are coupled in parallel with each other via the connecting portion **CP3**. As shown in FIG. 1A, most segments **L1** and **L2** are arranged between the first turn and the second turn of the coil **160**, and located between the third turn and the fourth turn of the coil **160**. In this example, the outermost turn of the coil **120** or **140** is arranged in the coil **160**. In other words, a range of each of the coils **120** and **140** partially overlaps a range of the coil **160**. With the above arrangements, the coil **160** is able to couple the coil **120** and the coil **140**. As a result, a signal received by the coil **160** may be simultaneously coupled to the coil **140** and the coil **120**, in order to output two sets of differential signals. Alternatively, differential signals received by the coil **140** and the coil **120** may be coupled to the coil **160**, in order to be combined as a single signal.

[0030] In some embodiments, the coil **120** may include additional connecting portions **CP1** (not shown) and additional vias (not shown), which may be configured to couple the segments **L1** in the

innermost turn. In some embodiments, as shown in FIG. 1A to FIG. 1C, the segments L1, L2, and L3 may be implemented with a first metal layer, the connecting portions CP1, CP2, and CP3 may be implemented with a second metal layer, and the first metal layer is different from the second metal layer. For example, the first metal layer may be (but not limited to) an ultra-thick metal (UTM) layer, and the second metal layer may be (but not limited to) a re-distribution layer (RDL). [0031] FIG. 2 is a schematic diagram of a transformer device 200 according to some embodiments of the present disclosure. Compared with the transformer device 100 in FIG. 1A, in this example, a number of crossing portions in the transformer device 200 is lower. Each of the coils 120 and 140 crosses the coil 160 twice in the first side of the coil 160, and crosses the coil 160 twice in the second side of the coil 160. In some embodiments, certain turns (e.g., the outermost turn and a second turn adjacent to the outermost turn) in both of the first winding (which may be, for example, formed with the segments L1 and the connecting portions CP1) and the second winding (which may be, for example, formed with the segments L2 and the connecting portions CP2) form crossing portions in the first side and the second side of the coil 160. In greater detail, in the first side, two connecting portions CP1 and CP2 cross the segments L3 in the outermost turn of the coil 160, and the corresponding segments L1 and L2 cross the connecting portion CP3 in the outermost turn of the coil 160. Compared with FIG. 1A, in this example, the coils 120 and 140 do not cross the third turn (or the innermost turn) of the coil 160 in the first side of the coil 160, and do not cross the third turn (or the innermost turn) of the coil 160 in the second side of the coil 160. In this example, the innermost turn and the third turn adjacent to the innermost turn in the each of the first winding and the second winding and the coil 160 do not form crossing portions in the first side and the second side.

[0032] In other words, in the first side and the second side, the third turn and the fourth turn in the coil 120 are directly formed with the segments L1 (i.e., without connecting portions CP1), and the third turn and the fourth turn in the coil 140 are directly formed with the segments L2 (i.e., without connecting portions CP2), and the third turn and the fourth turn in the coil 160 are directly formed with the segments L3 (i.e., without connecting portions CP3). As a result, the number of the crossing portion is lower. In different applications, the number of the crossing portions may be utilized to adjust an inductance ratio among the coils 120, 140, and 160. For example, if the number of the crossing portions is higher, the capacitance and the mutual inductance among the coils 120, 140, and 160 will be higher, which improves the mutual inductance among the coils 120, 140, and 160.

[0033] FIG. 3 is a schematic diagram of a transformer device 300 according to some embodiments of the present disclosure. Compared with the transformer device 100 in FIG. 1A or the transformer device 200 in FIG. 2, the coils 120, 140, and 160 in FIG. 3 have different number of turns.

[0034] In this example, the segments L1 and the connecting portions CP1 form a first winding having two turns. For example, from the terminal P11 to the terminal P12, the segments L1 and the connecting portions CP1 sequentially form about a quarter of the first turn (e.g., the outermost turn), about a quarter of the second turn (e.g., the innermost turn), about a quarter of the first turn, about a half of the second turn, about a quarter of the first turn, about a quarter of the second turn, and about a quarter of the first turn. Similarly, the segments L2 and the connecting portions CP2 form a second winding having two turns. For example, from the terminal P21 to the terminal P22, the segments L2 and the connecting portions CP2 sequentially form about a quarter of the first turn (e.g., the outermost turn), about a quarter of the second turn (e.g., the innermost turn), about a quarter of the first turn, about a half of the second turn, about the quarter of the first turn, about the quarter of the second turn, and about a quarter of the first turn.

[0035] In this example, the coil 120 further includes a terminal P13, which is a middle terminal between the terminal P11 and the terminal P12. In some embodiments, the terminal P13 may be a center tap terminal. The terminal P13 may be extended along the direction of -Y through an additional segment L1, in order to receive a common mode voltage (e.g., an AC ground voltage).

Similarly, the coil **140** further includes a terminal **P23**, which is a middle terminal between the terminal **P11** and the terminal **P22**. In some embodiments, the terminal **P23** may be a center tap terminal. The terminal **P23** may be extended along the direction of $-Y$ through an additional connecting portion **CP2**, in order to receive the common mode voltage.

[0036] In this example, the segments **L3** in the outermost turn and the innermost turn of the coil **160** are coupled in parallel with each other via the connecting portions **CP3**. A part of the segments **L1** and a part of the segments **L2** are arranged between the segments **L3** that are coupled in parallel with each other. For example, the segments **L1** in the second turn of the coil **120** and the segments **L2** in the second turn of the coil **140** are arranged between the segments **L3** that are coupled in parallel with each other. In this example, the range of each of the coils **120** and **140** partially overlaps the range of the coil **160**, and the outermost turn of the coil **120** or **140** is arranged outside the coil **160**. With the above arrangement, the coil **160** is able to couple the coils **120** and **140**.

[0037] Similar to FIG. 1A, in the first side (e.g., a side along the direction of $-X$) and the second side (e.g., a side along the direction of $+X$) of the coil **160**, the coil **160** and each turn in the first winding and the second winding form crossing portions. In FIG. 3, each of the coils **120** and **140** crosses the coil **160** twice in the first side of the coil **160**, and crosses the coil **160** twice in the second side of the coil **160**. For example, in the first side, the connecting portions **CP1** and **CP2** cross the segment **L3** in the innermost loop of the coil **160**, and the corresponding segments **L1** and **L2** cross the connecting portion **CP3** in the outermost turn of the coil **160**. Based on the symmetrical structure, each of the coils **120** and **140** crosses the coil **160** twice in the second side of the coil **160**. Similarly, each of the coils **120** and **140** crosses the coil **160** four times in the third side of the coil **160**, and crosses the coil **160** twice in the fourth side of the coil **160**, in which the third side and the fourth side are the other two opposite sides of the coil **160**. With the above arrangement, the coils **120**, **140**, and **160** may form crossing portions via the connecting portions **CP1**, **CP2**, and **CP3**, respectively. Similar to the transformer device **100** in FIG. 1A, the segments **L1**, **L2**, and **L3** may be twisted around each other via the connecting portions **CP1**, **CP2**, and **CP3**.

[0038] FIG. 4 is a schematic diagram of a transformer device **400** according to some embodiments of the present disclosure. Compared with the transformer device **300** in FIG. 3, in this example, a number of the crossing portions in the transformer device **400** is lower. In greater detail, each of the coil **120** and **140** does not cross the coil **160** in the first side and the second side of the coil **160**. In other words, in the first side and the second side, the first turn and the second turn in the coils **120**, **140**, **160** are directly formed with the segments **L1**, **L2**, and **L3**, respectively (i.e., without the connecting portions **CP1**, **CP2**, or **CP3**). As a result, the number of the crossing portions is lower, in order to achieve different inductance ratio.

[0039] FIG. 5 is a schematic diagram of a transformer device **500** according to some embodiments of the present disclosure. Although the shape and the entire area are different from the transformer device **100** in FIG. 1A, the arrangement of the transformer device **500** is similar to that of the transformer device **100**. Similar to the transformer device **100** in FIG. 1A or the transformer device in FIG. 2, each of the coils **120**, **140**, and **160** is a winding having four turns. Compared with the transformer device **100** in FIG. 1A, in this example, the coil **120** only includes one connecting portion **CP1**, and the coils **120**, **140**, and **160** are directly formed through the segments **L1**, **L2**, and **L3**, respectively, in the first side (e.g., the side along the direction of $-X$) and the second side (e.g., the direction of $+X$). In other words, compared with the transformer device **100**, the number of crossing portions in the transformer device **500** is lower.

[0040] In greater detail, from the terminal **P11** to the terminal **P12**, the segments **L1** and the connecting portions **CP1** sequentially form about a half of the first turn (e.g., the outermost turn), about a half of the second turn, about a half of the third turn, the fourth turn (e.g., the innermost turn), about a half of the third turn, about a half of the second turn, and about a half of the first turn. Similarly, from the terminal **P21** to the terminal **P22**, the segments **L2** and the connecting portions **CP2** sequentially form about a half of the first turn (e.g., the outermost turn), about a half of the

second turn, about a half of the third turn, the fourth turn (e.g., the innermost turn), about a half of the third turn, about a half of the second turn, and about a half of the first turn.

[0041] The segments **L3** in the first turn (e.g., the outermost turn) and the second turn of the coil **160** are coupled in parallel with each other via the connecting portions **CP3**. The segments **L3** in the third turn and the fourth turn (e.g., the innermost turn) of the coil **160** are coupled in parallel with each other via the connecting portions **CP3**. A first part of the segments **L1** and **L2** is arranged between the first turn and the second turn of the coil **160**, and a second part of the segments **L1** and **L2** is arranged between the third turn and the fourth turn of the coil **160**. The segments **L1** forming the outermost turn of the coil **120** and the segments **L2** forming the outermost turn of the coil **140** are arranged outside the outermost turn of the coil **160**. In this example, the range of the coil **160** partially overlaps the range of each of the coils **120** and **140**. With such arrangement, the coil **160** is able to couple the coils **120** and **140**.

[0042] On the other hand, in this example, the coils **120** and **140** cross the coil **160** six times in the third side of the coil **160**, and cross the coil **160** six times in the fourth side of the coil **160**, in which the third side and the fourth side are two opposite sides of the coil **160**. For example, the third side is a side along the direction of +Y, and the fourth side is a side along the direction of -Y. In greater detail, in the third side, the segments **L1** in the outermost turn of the coil **120** and the segments **L2** in the outermost turn of the coil **140** cross the connecting portions **CP3** between the first turn and the second turn of the coil **160**, and the connecting portions **CP1** in the outermost turn of the coil **120** and the connecting portions **CP2** in the outermost turn of the coil **140** cross the segments **L3** in the outermost turn of the coil **160**. In the third side, the segments **L1** in the third turn of the coil **120** and the segments **L2** in the fourth turn of the coil **140** cross the connecting portions **CP3** between the third turn and the fourth turn of the coil **160**. Similarly, in the fourth side, the segment **L1** in the second turn of the coil **120** and the segments **L2** in the second turn of the coil **140** cross the connecting portions **CP3** between the first turn and the second turn of the coil **160**, and the connecting portion **CP1** between the second turn and the third turn of the coil **120** and the connecting portion **CP2** between the second turn and the third turn of the coil **140** cross the segments **L3** in the third turn of the coil **160**. In the fourth side, the segments **L1** in the innermost turn of the coil **120** and the segments **L2** in the innermost turn of the coil **140** cross the connecting portions **CP3** between the third turn and the fourth turn of the coil **160**.

[0043] With above arrangements, the coils **120**, **140**, and **160** may form crossing portions via the connection portions **CP1**, **CP2**, and **CP3**, respectively. Equivalently, the segments **L1**, **L2**, and **L3** may be twisted around each other via the connecting portions **CP1**, **CP2**, and **CP3**.

[0044] The above implementations (e.g., a number of turns, materials, a number of terminals, shapes, etc.) and application examples about the above transformer devices **100**, **200**, **300**, **400**, and **500** are given for illustrative purposes, and the present disclosure is not limited thereto. For example, the shape of the coil **120**, the coil **140**, and the coil **160** may be other polygons or circle. The number of turns in the coil **120**, the coil **140**, and the coil **160**, or a wire distance between segments can be adjusted according to practical requirement.

[0045] As described above, the transformer device in some embodiments of the present disclosure may utilize three coils to implement an inductor structure having mirror-symmetry. As a result, the transformer device may achieve a better wire balance, in order to be applied to a power combiner, a balance-to-unbalance conversion, an unbalance-to-balance conversion, and so on. Moreover, according to different applications, the transformer device may utilize twisted wires to achieve uniform coupling.

[0046] The aforementioned descriptions represent merely some embodiments of the present disclosure, without any intention to limit the scope of the present disclosure thereto. Various equivalent changes, alterations, or modifications based on the claims of present disclosure are all consequently viewed as being embraced by the scope of the present disclosure.

Claims

1. A transformer device, comprising: a first coil comprising a plurality of first segments and at least one first connecting portion, wherein the plurality of first segments are coupled to each other through the at least one first connecting portion; a second coil comprising a plurality of second segments and a plurality of second connecting portions, wherein the plurality of second segments are coupled to each other through the plurality of second connecting portions; and a third coil configured to couple the first coil and the second coil, wherein the third coil comprises a plurality of third segments and a plurality of third connecting portions, a part of the plurality of third segments are coupled in parallel with each other through the plurality of third connecting portions, and at least one part of the plurality of first segments and at least one part of the plurality of second segments are arranged between the part of the plurality of third segments, wherein the plurality of first segments and the at least one first connecting portion form a first winding, the plurality of second segments and the plurality of second connecting portions form a second winding, and the third coil and an innermost turn and a turn adjacent to the innermost turn in each of the first winding and the second winding do not form a plurality of crossing portions in two opposite sides of the third coil.
2. The transformer device of claim 1, wherein the plurality of first segments, the plurality of second segments, and the plurality of third segments are twisted around each other through the at least one first connecting portion, the plurality of second connecting portions, and the plurality of third connecting portions.
3. The transformer device of claim 1, wherein an outer turn and a turn adjacent to the outer turn in each of the first winding and the second winding form the plurality of crossing portions in the two opposite sides of the third coil.
4. The transformer device of claim 3, wherein the two opposite sides are a first side and a second side of the third coil, and the plurality of crossing portions in the first side are symmetric to the plurality of crossing portions in the second side.
5. The transformer device of claim 1, wherein the two opposite sides are a first side and a second side of the third coil, and each of the first coil and the second coil crosses the third coil twice in the first side, and crosses the third coil twice in the second side.
6. The transformer device of claim 1, wherein each of the first coil and the second coil comprises a plurality of terminals, and the plurality of terminals are not arranged in the two opposite sides.
7. The transformer device of claim 1, wherein a range of each of the first coil and the second coil partially overlaps a range of the third coil.
8. The transformer device of claim 1, wherein an outermost turn of the first coil or the second coil is arranged in the third coil.
9. The transformer device of claim 1, wherein each of the first coil, the second coil, and the third coil substantially has a mirror-image symmetry.
10. The transformer device of claim 1, wherein the plurality of first segments, the plurality of second segments, and the plurality of third segments are implemented with a first metal layer, and the at least one first connecting portion, the plurality of second connecting portions, and the plurality of third connecting portions are implemented with a second metal layer, and the first metal layer is different from the second metal layer.
11. The transformer device of claim 1, wherein the first winding has four turns, and the second winding has four turns.
12. The transformer device of claim 1, wherein the plurality of first segments, the plurality of second segments, and the plurality of third segments are twisted around each other via the at least one first connecting portion, the plurality of second connecting portions, and the plurality of third connecting portions.

- 13.** A transformer device, comprising: a first coil comprising a plurality of first segments and at least one first connecting portion, wherein the plurality of first segments are coupled to each other through the at least one first connecting portion; a second coil comprising a plurality of second segments and a plurality of second connecting portions, wherein the plurality of second segments are coupled to each other through the plurality of second connecting portions; and a third coil configured to couple the first coil and the second coil, wherein the third coil comprises a plurality of third segments and a plurality of third connecting portions, a part of the plurality of third segments are coupled in parallel with each other through the plurality of third connecting portions, and at least one part of the plurality of first segments and at least one part of the plurality of second segments are arranged between the part of the plurality of third segments, wherein the plurality of first segments and the at least one first connecting portion form a winding having a first turn and a second turn, and the plurality of first segments and the at least one first connecting portion sequentially form, from a first terminal of the first coil to a second terminal of the first coil, a quarter of the first turn, a quarter of the second turn, a quarter of the first turn, a half of the second turn, a quarter of the first turn, a quarter of the second turn, and a quarter of the first turn.
- 14.** The transformer device of claim 13, wherein an outermost turn of the first coil or the second coil is arranged outside the third coil.
- 15.** The transformer device of claim 13, wherein a range of each of the first coil and the second coil partially overlaps a range of the third coil.
- 16.** The transformer device of claim 13, wherein each of the first coil, the second coil, and the third coil substantially has a mirror-image symmetry.
- 17.** The transformer device of claim 13, wherein the plurality of first segments, the plurality of second segments, and the plurality of third segments are implemented with a first metal layer, and the at least one first connecting portion, the plurality of second connecting portions, and the plurality of third connecting portions are implemented with a second metal layer, and the first metal layer is different from the second metal layer.
- 18.** The transformer device of claim 13, wherein each of the first coil and the second coil crosses the third coil twice in a first side of the third coil, crosses the third coil twice in a second side of the third coil, crosses the third coil four times in a third side of the third coil, crosses the third coil twice in a fourth side of the third coil, the first side and the second side are two opposite sides, and the third side and the fourth side are other two opposite sides.
- 19.** The transformer device of claim 13, wherein the plurality of first segments, the plurality of second segments, and the plurality of third segments are twisted around each other via the at least one first connecting portion, the plurality of second connecting portions, and the plurality of third connecting portions.
- 20.** The transformer device of claim 13, wherein the first coil further comprises a third terminal, and the third terminal is a middle terminal between the first terminal and the second terminal of the first coil, and is configured to receive a common mode voltage.
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